Advances in Cognitive Systems X (20XX) 1-6 Submitted X/20XX; published X/20XX

Diffusion Models and Concept Formation

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Abstract

Humans organize knowledge into a taxonomy of concepts with nested levels of abstraction and a *basic level* at which people recognize and name objects with the least cognitive effort. Cobweb is a classic cognitive account of this ability, an incremental learner that builds a probabilistic concept hierarchy by maximizing category utility. We argue that diffusion models, although designed for image synthesis, implicitly perform the same computation. The noisy marginals of a diffusion model are Gaussian smoothings of the data distribution, and the modes of these marginals form a hierarchy that corresponds to a Cobweb tree of probabilistic prototypes in four respects. Both are hierarchical density models, both are hierarchical-Bayesian models with Gaussian prototypes, both treat categorization as score-following that reduces uncertainty, and in both a basic level emerges. We locate this basic level for a diffusion model at an intermediate noise level, where recent analyses show that the reverse process commits to the class identity of a sample. The two models differ mainly in how they represent and learn the taxonomy. Cobweb learns a discrete tree incrementally, whereas a diffusion model encodes a continuous, interpolable hierarchy in a single learned score field fit to the data distribution. We test the correspondence on MNIST and Fashion-MNIST by recovering the diffusion hierarchy through mode-finding and comparing the basic levels of the two models. This reframes diffusion as a cognitive model of concept formation and offers Cobweb a continuous, scalable instantiation.

1. Introduction

Humans naturally organize their knowledge into nested levels of abstraction. A collie is a kind of dog, a dog a kind of mammal, a mammal a kind of animal. Within these hierarchies, one level supports recognition with the least cognitive effort. This *basic level* (e.g., *dog*, *chair*) is the level people use by default, because it offers the best trade-off between being informative about an instance's features and being distinctive from other categories (Rosch et al., 1976; Rosch & Mervis, 1975). Basic categories maximize the joint predictability of cues and category membership (Jones, 1983), and a computational account of human categorization should explain why such a level emerges.

Cobweb (Fisher, 1987) offers a classic computational account. It processes instances one at a time and incrementally builds a probabilistic concept hierarchy, choosing at each step the restructuring operation that most improves *category utility*, an information-theoretic measure of how much a partition increases the predictability of instance features (Corter & Gluck, 1992). The resulting tree reproduces basic-level and typicality effects from the psychology of categorization (Fisher, 1988).

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Recent work has scaled the framework well beyond its original nominal-attribute setting, to visual classification without catastrophic forgetting (Barari et al., 2024), to semantic retrieval (Gupta et al., 2025), to masked language modeling (Lian et al., 2025), and to deep taxonomic networks, which learn a hierarchy of Gaussian prototypes in a latent space (Wang et al., 2025a).

In parallel, and largely independently, diffusion models (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021) have become a leading approach to image generation. A diffusion model learns to reverse a gradual noising process by estimating the *score* (the gradient of the log-density) of the data corrupted at many noise levels. These models are typically studied as black-box samplers and evaluated on sample quality, not as models of human concept formation. Yet the object they learn, a family of progressively smoothed data densities, has the structure of a concept hierarchy, a continuum of representations of the data at different levels of abstraction.

This paper argues that the two lines of work describe members of the same family of models. Our contributions are as follows.

- We show that diffusion models and Cobweb belong to the same family of hierarchical Bayesian density models with Gaussian prototypes, by making four formal correspondences precise (Section 3).
- We argue that diffusion models have a *basic level*, an intermediate noise level at which concepts are recognized with the least cognitive effort.
- We validate our claim by showing that hierarchies recovered through score-based mode-finding are consistent with Cobweb trees, and that their basic levels align with Cobweb's, with results on MNIST and Fashion-MNIST.

The remainder of the paper reviews Cobweb, diffusion models, and hierarchical generative clustering (Section 2), develops their correspondences (Section 3), presents the empirical validation (Section 4), and discusses implications (Section 5).

2. Background

2.1 Concept Formation with Cobweb

Cobweb (Fisher, 1987) is an incremental and unsupervised model of human concept formation. It processes a stream of instances and builds a hierarchy of probabilistic concepts, where each concept node maintains a summary of the instances it has adopted. As each instance arrives, Cobweb sorts it down the tree from the root, and at every node it considers four operations. It can add the instance to the best-fitting child, create a new child for it, merge the two best children into one, or split the best child and promote its children. It applies whichever operation most improves category utility. In the original model each instance is a set of nominal attribute–value pairs and a node stores a table of attribute probabilities. Cobweb/3 extends the framework to continuous attributes by storing, for each attribute, the mean and variance of a Gaussian (McKusick & Thompson, 1990). We therefore take a node c to carry a prior $P(c)$ and a diagonal Gaussian over features, $p(x \mid c) = \mathcal{N}(x; \mu_c, \sigma_c^2 I)$, treating features as conditionally independent given the concept.

The measure that drives every decision is category utility. In its information-theoretic form (Corter & Gluck, 1992), adopted by Cobweb/4V (Barari et al., 2024) and by deep taxonomic net-

works (Wang et al., 2025a), it scores a partition $\{C_1, \ldots, C_n\}$ of a node's instances by the expected reduction in the uncertainty of feature values that knowing the child provides,

$$\mathrm{CU}(\{C_1, \ldots, C_n\}) = \sum_{k=1}^{n} P(C_k) \left[H(A) - H(A \mid C_k) \right], \tag{1}$$

where $H(A) = -\sum_i \sum_j P(A_i{=}V_{ij}) \log P(A_i{=}V_{ij})$ is the entropy over attribute values. Expressing the measure in terms of entropy is what allows arbitrary distributions, such as Gaussians, to be plugged in, since one only needs a closed-form entropy. Writing X for the random variable collecting the attributes and C for the concept an instance is assigned to, category utility is precisely the mutual information between features and concept (Wang et al., 2025a),

$$\mathrm{CU} = I(X; C) = H(X) - \sum_{k=1}^{n} P(C_k)\, H(X \mid C_k). \tag{2}$$

A partition has high category utility when its clusters have low internal entropy (predictable features) yet collectively account for the variation in the parent (distinctiveness).

Although it was introduced decades ago (Fisher, 1987; Gennari et al., 1989) and has long been studied as a model of human categorization (Iba & Langley, 2011), Cobweb has seen renewed development, much of it aimed at scaling the framework past nominal attributes and small symbolic datasets. Cobweb/4V learns concept hierarchies over images and resists catastrophic forgetting in continual learning, without replay buffers or regularization (Barari et al., 2024), and a follow-up traces this robustness to its closed-form, information-theoretic updates rather than to gradient descent (Barari et al., 2025). Concept-formation variants induce language models and perform masked word prediction at scale (MacLellan et al., 2022; Lian et al., 2025), a Cobweb tree over sentence embeddings supports coarse-to-fine semantic retrieval (Gupta et al., 2025), and, as a cognitive model, Cobweb accounts for human-like incremental category learning (Lian et al., 2024). Most relevant here, recent work connects Cobweb with neural learning. Taxonomic networks treat the concept hierarchy as a shared representation for pairing symbolic and neural systems (Wang et al., 2025b), and deep taxonomic networks recast it as a deep generative model, learning a tree-structured mixture-of-Gaussians prior whose training objective coincides with category utility (Wang et al., 2025a). The latter connects Cobweb directly to the generative latent-variable models we review in Section 2.3.

2.2 Diffusion Models

A diffusion model is defined by a forward process that gradually corrupts data with Gaussian noise and a learned reverse process that removes it (Ho et al., 2020; Song et al., 2021). The forward process sets $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ and a decreasing schedule $\{\bar{\alpha}_t\}$, so that the distribution of x_t is the data distribution convolved with a Gaussian kernel, a kernel density estimate of the data at bandwidth $\sigma_t := \sqrt{1 - \bar{\alpha}_t}$,

$$p_t(x_t) = \int p_{\text{data}}(x_0)\, \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\, x_0, (1 - \bar{\alpha}_t) I\big)\, dx_0. \tag{3}$$

The amount of corruption is summarized by the signal-to-noise ratio, $\mathrm{SNR}(t) = \bar{\alpha}_t/(1-\bar{\alpha}_t)$, which decreases monotonically from high values (almost-clean data, fine detail visible) at small t to near zero (almost pure noise) at large t (Kingma et al., 2021).

Rather than learn p_t directly, a diffusion model learns its *score*, the gradient of the log-density, which a standard result identifies with a rescaled denoiser (Hyvärinen, 2005; Vincent, 2011; Song et al., 2021),

$$s_\theta(x_t, t) \approx \nabla_{x_t} \log p_t(x_t) = -\frac{\varepsilon_\theta(x_t, t)}{\sqrt{1-\bar{\alpha}_t}}. \tag{4}$$

The score also gives, through Tweedie's empirical-Bayes formula (Robbins, 1956; Efron, 2011), the posterior mean of the clean datum given its corrupted version,

$$\hat{x}_0(x_t) = \mathbb{E}[x_0 \mid x_t] = \frac{1}{\sqrt{\bar{\alpha}_t}}\big(x_t + (1-\bar{\alpha}_t)\,\nabla_{x_t} \log p_t(x_t)\big). \tag{5}$$

A local mode of p_t has vanishing score, so by Eq. 5 its rescaled position lies on the clean data manifold, $\hat{x}_0(x_t^\star) = x_t^\star/\sqrt{\bar{\alpha}_t}$ (Wang et al., 2026). We recover such a mode from any datum x_0 by noising it to level t and then ascending the score until it converges to a fixed point,

$$x_t^{(0)} = \sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon, \qquad x_t^{(i+1)} = x_t^{(i)} + \eta\, s_\theta\big(x_t^{(i)}, t\big). \tag{6}$$

The noise level t at which we do this fixes the level of abstraction of the concept the datum is mapped to. A small t returns a near-instance prototype, and a large t returns a broad class prototype. A diffusion model thus provides a smoothed density at every noise level, together with a score that locates the modes of that density and their local Gaussian shape. These are the ingredients we need to connect it to Cobweb.

Throughout the paper we keep two indices distinct. The noise level is the diffusion index t, whose smoothing bandwidth $\sigma_t = \sqrt{1-\bar{\alpha}_t}$ grows with t, and the level of abstraction is a position in the concept hierarchy that runs from fine to coarse. A higher noise level gives a larger bandwidth, and a larger bandwidth gives a coarser level of abstraction. We use *noise level* for the diffusion index and *level of abstraction* for the hierarchy.

2.3 Hierarchical Clustering

A parallel and largely separate literature builds hierarchies with generative latent-variable models. The variational autoencoder learns a latent representation of the data by maximizing a variational lower bound on the likelihood (Kingma & Welling, 2014). Placing a clustering prior on that latent space turns it into a generative clustering model. VaDE uses a Gaussian-mixture prior to recover flat clusters (Jiang et al., 2017). Making the prior hierarchical yields a tree of clusters rather than a flat partition. TreeVAE grows a latent tree whose leaves are clusters (Manduchi et al., 2023), DeepECT embeds a cluster tree inside an autoencoder (Mautz et al., 2020), and, in the Bayesian-nonparametric tradition, the nested Chinese restaurant process places a prior over trees of unbounded depth (Blei et al., 2010). Most relevant here, the deep taxonomic network learns a binary-tree mixture-of-Gaussians prior and shows that maximizing its evidence lower bound is equivalent to maximizing Cobweb's category utility (Wang et al., 2025a), placing Cobweb and deep generative clustering on a common footing.

Table 1. The correspondence between Cobweb concept hierarchies and the implicit hierarchy of a diffusion model. Each row is argued in the indicated subsection of Section 3.

Cobweb		**Diffusion model**
tree depth (level of abstraction)	$\Longleftrightarrow$	noise level σ_t, equivalently $\mathrm{SNR}(t)$ (§3.1)
node Gaussian $\mathcal{N}(\mu_c, \sigma_c^2 I)$	$\Longleftrightarrow$	mode Gaussian $\mathcal{N}(m_c, \Sigma_c)$ (§3.2)
root-to-leaf categorization	$\Longleftrightarrow$	coarse-to-fine denoising (§3.3)
basic level $\arg\max_{c \in \pi(x)} \mathcal{D}(c)$	$\Longleftrightarrow$	maximum-likelihood mode at noise $t^\star$ (§3.4)

These models inherit a long tradition. Classical hierarchical clustering builds a tree agglomeratively, merging the pair of clusters that least increases within-cluster variance (Ward, 1963), or divisively, by splitting from the top down. Density-based methods instead treat clusters as the modes of a density and assign points to their basins of attraction, as in mean shift (Comaniciu & Meer, 2002) and density cluster trees. Cobweb is the cognitive-science member of this family, an incremental, divisive conceptual clustering that summarizes each node with a probabilistic prototype and chooses operations by category utility (Fisher, 1987). The deep latent-variable models extend this tradition by learning the representation and the hierarchy jointly, rather than clustering over fixed features.

We will argue that a diffusion model belongs to the same family, although it represents the hierarchy implicitly rather than storing a tree. Its noisy marginals are smoothed densities whose modes are cluster centroids at a continuum of noise levels, recovered by mode-seeking on the learned score, which is exactly the density-based view above. Two recent threads support this reading. Work on concept discovery uses the modes of a diffusion model's marginals directly as prototypes and composes them to generate new concepts (Wang et al., 2026; Du et al., 2020; Liu et al., 2022). And analyses of the diffusion process show that the structure it recovers across noise levels is genuinely hierarchical, with a phase transition at an intermediate noise level where coarse class identity is fixed while fine detail is not (Sclocchi et al., 2025; Biroli et al., 2024; Ambrogioni, 2025). These threads have developed separately from Cobweb. Our contribution is to draw the correspondence between them explicitly and to locate a basic level in the diffusion hierarchy.

3. Diffusion Models as Hierarchical Concept Formation

We fix a common notation before drawing the correspondences. Throughout, $x \in \mathbb{R}^d$ is a data point and X the random variable it realizes, $p_{\text{data}}(x_0)$ is the data distribution, c is a concept (a Cobweb node or a diffusion mode), and C is the concept an instance is assigned to. On the Cobweb side a concept is a Gaussian $\mathcal{N}(\mu_c, \sigma_c^2 I)$ at a node of a discrete tree. On the diffusion side a concept is a local mode $x_t^\star$ of a smoothed marginal p_t, equipped with a local Gaussian $\mathcal{N}(m_c, \Sigma_c)$ read off the score, as the next subsections develop. The four correspondences are summarized in Table 1 and argued in turn below. A final subsection discusses where the two models differ.

3.1 Both Are Hierarchical Density Models

A Cobweb tree is a hierarchy of densities. Each node summarizes its subtree with a probabilistic prototype, leaves capture individual instances, and ancestors describe progressively broader regions of feature space (Fisher, 1987). Because Cobweb sorts instances from the top down using its insert, create, merge, and split operators, it can be read as a divisive procedure that refines coarse regions of feature space into finer ones, keeping every node a good summary of the instances below it.

A diffusion model produces the same kind of hierarchy, indexed by a continuous noise level rather than by tree depth. For a finite dataset $\{x_0^{(i)}\}_{i=1}^N$, Eq. 3 is a Gaussian mixture with one component per training point,

$$p_t(x_t) = \frac{1}{N}\sum_{i=1}^{N} \mathcal{N}\big(x_t; \sqrt{\bar{\alpha}_t}\, x_0^{(i)}, (1-\bar{\alpha}_t)I\big), \tag{7}$$

a kernel density estimate whose bandwidth σ_t grows with t. The number of modes of this estimate falls monotonically as σ_t increases. Fine, instance-level modes at small t give way to broad, class-level modes at large t. These progressively smoothed densities, one at each noise level, form a family whose modes merge into a tree. A coarse mode at level t' is the parent of the finer modes at $t < t'$ that flow into it under score ascent, so depth in the tree corresponds to the bandwidth σ_t. Finding the modes at a fixed noise level is the mode-seeking problem that mean shift solves (Comaniciu & Meer, 2002), here driven by the learned score (Eq. 4) rather than a fixed kernel, and following modes across noise levels yields a cluster hierarchy in the spirit of agglomerative clustering (Ward, 1963) but read off a single learned density. Empirically, modes of p_t consolidate from instance-level to object-level prototypes as t grows (Wang et al., 2026), and the structure that diffusion recovers across noise levels is hierarchical (Sclocchi et al., 2025), closely matching the tree Cobweb learns on the same data (see Figure 1).

3.2 Both Are Hierarchical Bayes with Gaussian Prototypes

The prototypes in both hierarchies are Gaussian, and in both the broader concepts are mixtures of the narrower ones. A Cobweb node stores a diagonal Gaussian $\mathcal{N}(\mu_c, \sigma_c^2 I)$ that pools the statistics of its children. With $P(k \mid c)$ the weight of child k under parent c, the parent mean is the convex combination of the children's means,

$$\mu_c = \sum_k P(k \mid c)\, \mu_k, \qquad \sum_k P(k \mid c) = 1, \tag{8}$$

and the law of total variance gives $\sigma_c^2 = \sum_k P(k \mid c)\,[\,\sigma_k^2 + (\mu_k - \mu_c)^2\,]$, so a parent is a broader Gaussian that subsumes its children and the hierarchy is a tree-structured mixture of Gaussians (McKusick & Thompson, 1990). Deep taxonomic networks impose a binary tree and learn the same pooling as a single mixing weight, $\mu_c = \lambda\, \mu_{\text{left}} + (1-\lambda)\, \mu_{\text{right}}$ with $\lambda \in [0,1]$, the two-child special case of Eq. 8 (Wang et al., 2025a). A diffusion model gives the same object analytically. At a mode $x_t^\star$, Tweedie's formula (Eq. 5) places the prototype mean at $m_c = x_t^\star/\sqrt{\bar{\alpha}_t}$, and the Jacobian of the denoiser sets its covariance,

$$\Sigma_c = \frac{1-\bar{\alpha}_t}{\sqrt{\bar{\alpha}_t}}\, \nabla_{x_t} \hat{x}_0(x_t)\big|_{x_t = x_t^\star}, \tag{9}$$

the posterior covariance of the clean datum, which at a mode is the Laplace approximation of p_t (Wang et al., 2026). Each mode thus carries a local Gaussian $\mathcal{N}(m_c, \Sigma_c)$, and the empirical marginal of Eq. 7 is a mixture of such Gaussians whose components broaden and fuse as t grows. The node Gaussians of Cobweb and the mode Gaussians of a diffusion model therefore play the same role. Both models explain a datum with a Gaussian component whose breadth encodes its level of abstraction (McKusick & Thompson, 1990; Wang et al., 2026).

3.3 Both Maximize Information About the Data

The third correspondence is that both models pursue the same objective, making a concept as informative as possible about the data it explains, even though they pursue it by different mechanisms. In Cobweb the objective is category utility, which by Eq. 2 is the mutual information $I(X; C)$ between the features and the concept. The feature entropy $H(X)$ is fixed, so maximizing $I(X; C)$ minimizes the residual uncertainty $H(X \mid C)$, and a concept scores well when it leaves little to guess about its members. A diffusion model optimizes the matching quantity for a continuous hierarchy. Its training objective is a denoising error weighted by the rate of change of the signal-to-noise ratio (Kingma et al., 2021),

$$\mathcal{L}(\theta) = \tfrac{1}{2}\,\mathbb{E}_{x_0,\varepsilon} \int_0^1 w(t)\,\left\|x_0 - \hat{x}_\theta(x_t, t)\right\|^2 dt, \qquad w(t) = -\,\mathrm{SNR}'(t), \tag{10}$$

with $\hat{x}_\theta$ the denoiser of Eq. 5 and $w(t) \geq 0$ because the ratio decreases in t. By the I-MMSE relation (Guo et al., 2005), the best achievable value of this integrated error is the mutual information the noisy observations carry about the clean datum, and at the optimal denoiser it equals the datum's negative log-likelihood exactly (Kong et al., 2023). Minimizing Eq. 10 therefore drives the model to extract as much information about the clean datum as the noise leaves available, at every noise level. Category utility and the denoising objective are two forms of the same principle, each making the concept retain as much information about the data as it can.

The two objectives are pursued by different search procedures, and this is where the coarse-to-fine order appears. Cobweb sorts an instance from the root down, at each node applying whichever of its four operators yields the child partition of highest category utility, so it commits to broad category structure first and refines toward the leaf. A diffusion model reaches the same order by sampling, following the learned score from low signal-to-noise (coarse, class-level structure) to high signal-to-noise (fine, instance-level detail) (Song & Ermon, 2019; Karras et al., 2022). The score points in the direction the trained denoiser estimates, so each step refines a coarse guess into a more specific one, the continuous analog of Cobweb's root-to-leaf sort. Both models therefore share this objective and this coarse-to-fine order, and differ in how they represent and search the hierarchy, a difference we take up in Section 3.5.

3.4 A Basic Level of Greatest Distinctiveness

The basic level is the level of abstraction at which a learner obtains the useful category most informative about its members, distinctive enough to set them apart from the data as a whole, yet not so specific that it collapses onto individual instances (Rosch et al., 1976; Fisher, 1988). On the cognitive science account the basic level is a frontier of concepts across the hierarchy (Fisher, 1988).

Table 2. Distinctiveness $\mathcal{D}(c)$, the quantity that locates the basic level, shown with its pointwise ingredient $\mathrm{pmi}(x; c)$ and its dataset-level aggregate $I(X; C)$. Distinctiveness is the expected pointwise mutual information of a concept's members. The pointwise term is signed, and $\mathcal{D}(c)$ and the mutual information are nonnegative.

Scope	**Quantity**	**Reading**
specific x and c	$\mathrm{pmi}(x; c) = \log \dfrac{p(x \mid c)}{p(x)}$	evidence that x belongs to c
fixed c, over its members x	$\mathcal{D}(c) = \mathbb{E}_{x \sim p(x \mid c)}[\mathrm{pmi}]$	distinctiveness of a concept, the KL specific surprise
over both c and x	$I(X; C) = \sum_c p(c)\, \mathcal{D}(c)$	mutual information, the counterpart of category utility

These concepts need not sit at a uniform depth, and each is the concept of greatest distinctiveness along its path, the one most informative about its own members. We measure how well a concept meets this standard by its distinctiveness,

$$\mathcal{D}(c) = \mathbb{E}_{x \sim p(x \mid c)}\big[\mathrm{pmi}(x; c)\big] = D_{\mathrm{KL}}\big(p(x \mid c) \,\|\, p(x)\big) \geq 0, \tag{11}$$

the expected pointwise mutual information of its members, which is the Kullback–Leibler divergence of the concept from the data marginal and is what information theory calls the *specific surprise* of c (DeWeese & Meister, 1999). Two related quantities place $\mathcal{D}(c)$ in context (Table 2). Its pointwise ingredient, the pointwise mutual information $\mathrm{pmi}(x; c) = \log\big[p(x \mid c)/p(x)\big]$ of one datum and one concept, is the evidence that x belongs to c. Averaging $\mathcal{D}(c)$ in turn over concepts recovers the mutual information between data and concepts,

$$I(X; C) = \sum_c p(c)\, \mathcal{D}(c) \geq 0, \tag{12}$$

the information-theoretic counterpart of category utility, with the sum running over the concepts of one partition of the data. Distinctiveness is the per-concept quantity between these two, and it is the one we use to locate the basic level, by the same computation in Cobweb and in the diffusion model.

Distinctiveness is defined at every concept, so a basic level exists in any concept-formation system as the frontier where $\mathcal{D}$ peaks. Along each root-to-leaf path $\pi(x)$, the concepts that contain x from the root down to its leaf, we take the concept of greatest distinctiveness,

$$c_{\mathrm{basic}}(x) = \arg\max_{c \in \pi(x)} \mathcal{D}(c). \tag{13}$$

This score is converging to zero near the root, where $p(x \mid c) = p(x)$. It need not be monotone in depth, and in our experiments it reaches its maximum at an intermediate level of abstraction rather

than at the root or the leaves (Figure 2). In Cobweb the concepts are explicit, and we read this level directly off the learned tree (see Figure 3 for the basic-level prototypes of both models).

In a diffusion model the concepts are implicit. There is no stored tree and no closed form for $p(x \mid c)$, so the same level must be recovered empirically, by passing held-out data through the trained model. Each datum x induces a categorization path $\pi(x)$, the sequence of local modes from high noise (coarse) to low noise (fine), each carrying the local Gaussian $\mathcal{N}(m_c, \Sigma_c)$ of Section 3.2. For a fixed datum, the marginal $\log p(x)$ is constant across the concepts on its path. Thus, choosing the concept of greatest pointwise mutual information is the same as choosing the concept of greatest likelihood,

$$c^\star(x) = \arg\max_{c \in \pi(x)} \mathrm{pmi}(x; c) = \arg\max_{c \in \pi(x)} \big[\log p(x \mid c) - \log p(x)\big] = \arg\max_{c \in \pi(x)} \log p(x \mid c). \quad (14)$$

With the local covariance taken to be diagonal, $\Sigma_c = \mathrm{diag}(\sigma_{c,1}^2, \ldots, \sigma_{c,d}^2)$, the likelihood decomposes into a sum over coordinates of a standardized data-fit term and a log-variance term,

$$\log p(x \mid c) = -\frac{1}{2} \sum_{j=1}^{d} \frac{(x_j - m_{c,j})^2}{\sigma_{c,j}^2} \; - \; \frac{1}{2} \sum_{j=1}^{d} \log\big(2\pi\, \sigma_{c,j}^2\big). \quad (15)$$

The first term favors concepts whose centroids lie close to x relative to their spread. The second favors compact, low-variance concepts. Along a hierarchical path, these competing pressures can make the likelihood peak at an intermediate level of abstraction. We estimate $\mathcal{D}(c)$ in diffusion models from data by replacing the expectation in Eq. 11 with an empirical mean over the members assigned to a concept:

$$\mathcal{D}(c) \;\approx\; \frac{1}{N_c} \sum_{x \,\in\, \mathrm{members}(c)} \mathrm{pmi}(x; c), \qquad N_c = |\mathrm{members}(c)|. \quad (16)$$

Thus, $\mathcal{D}(c)$ can be measured from both models. In Cobweb we evaluate it at each node of the learned tree, approximating the marginal $p(x)$ by a mixture over the nodes. In the diffusion model we never form $p(x)$. The marginal cancels for a fixed datum in Eq. 14, and once we average over the held-out set it adds only a constant that is the same at every level, so the level at which mean distinctiveness peaks is the level at which the mean held-out log-likelihood peaks. This interior peak is a bias–variance tradeoff over levels of abstraction: the finest concepts are nearly unbiased for the instances they store but have high variance, predicting a held-out point poorly whenever it does not fall near one of them, whereas coarser concepts are more biased, pulled toward the mean of their members, yet vary less from one datum to the next. The basic level is the abstraction that balances the two. Read as a function of the level of abstraction, which is tree depth in Cobweb and noise level in the diffusion model, $\mathcal{D}$ is zero at the root and reaches its maximum at an intermediate level. The noise level $t^\star$ at which the diffusion curve peaks is its basic level (see Figure 2).

3.5 Discrete versus Continuous Hierarchies

The two models are not identical, however, and their differences are instructive. Cobweb learns an explicit, discrete tree, incrementally and one instance at a time, using symbolic insert, create, merge,

and split operators, and it stores one prototype per node. A diffusion model encodes a continuous hierarchy implicitly in a single learned score field. There is no stored tree, only a family of smoothed densities whose modes and their merging behavior define a hierarchy that must be recovered by mode-finding. One consequence is that diffusion prototypes are interpolable. A convex combination of two mode means, $m(\lambda) = \lambda\, m_a + (1 - \lambda)\, m_b$, traces a path of intermediate prototypes, whereas Cobweb's nodes are fixed and discrete. The trade-off is the familiar one between symbolic and sub-symbolic representations. Cobweb offers interpretability, resistance to catastrophic forgetting through sparse and local updates (Barari et al., 2025), and genuine incremental learning. A diffusion model offers continuity, scalability, and sample quality (Karras et al., 2022). Because the two describe the same structure by different means, a natural direction is to bridge them, for instance by pairing a neural density model with a symbolic taxonomy (Wang et al., 2025b; Gupta et al., 2025).

4. Validating the Connection

The correspondence of Section 3 makes empirical predictions. We lay them out as two questions, each comparing a diffusion model's implicit hierarchy, recovered by score-based mode-finding, against a Cobweb hierarchy learned on the same data. We use image benchmarks shared by both lineages. MNIST and Fashion-MNIST each contain 70,000 grayscale images of size 28×28 in ten classes, handwritten digits and clothing items respectively, split into 60,000 training and 10,000 test images. Both models are fit on the training images, with the Cobweb implementation following Cobweb/4V (Barari et al., 2024). Cobweb's concepts are explicit, so we read its hierarchy and basic level from the learned tree, whereas the diffusion model's concepts are implicit and we recover its hierarchy and basic level by passing the held-out test images through the trained model. The figures below report results on MNIST and Fashion-MNIST.

4.1 Does Mode-Finding Recover a Cobweb-Like Hierarchy?

The first prediction is that the hierarchy implicit in a diffusion model is similar to the one Cobweb learns. We recover the diffusion hierarchy bottom-up. Starting at a low noise level, we map every held-out test point to a fine mode by noising it to that level and ascending the score (Eq. 6), and group modes that fall within a variance threshold, taking each group's centroid to be the average of its modes. Raising the noise level in steps, we agglomerate these groups into parents, so that every parent encompasses the children beneath it and its centroid is the average of theirs (Section 3.2). Stacking the levels gives a taxonomy whose leaves are individual instances and whose root is a single coarse mode, which we compare to a Cobweb tree on the same data (Figure 1). Qualitatively, we see that both Cobweb and diffusion models create hierarchies in similar ways, merging similar stimuli (such as the 2s and 8s in the MNIST dataset, or the shoes and bags in the FashionMNIST dataset).

4.2 Does the Diffusion Basic Level Match Cobweb's?

The second prediction follows from Section 3.4. We score every concept by its distinctiveness $\mathcal{D}$, estimated as the average pointwise mutual information over the instances assigned to it (Eq. 16),

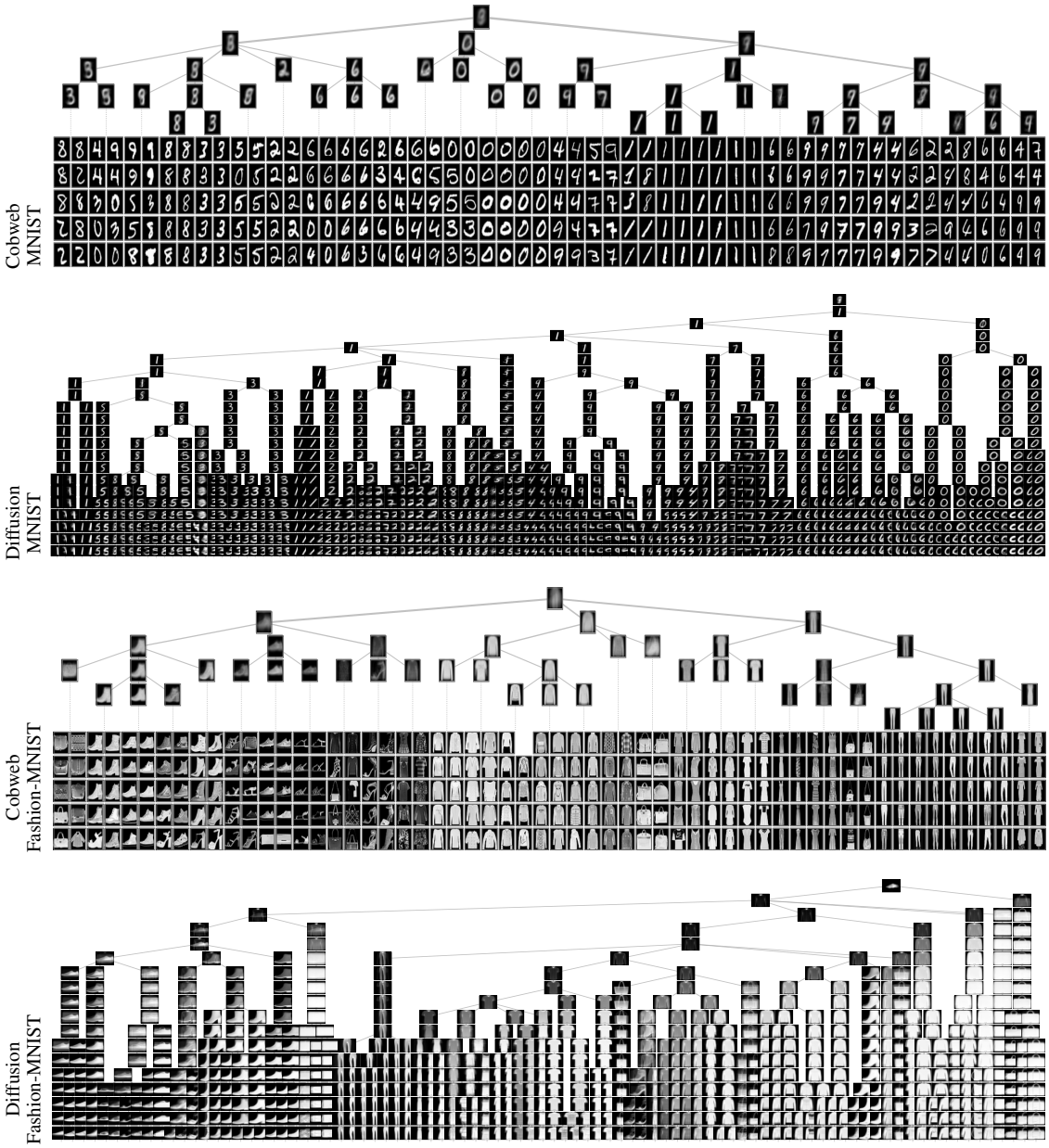
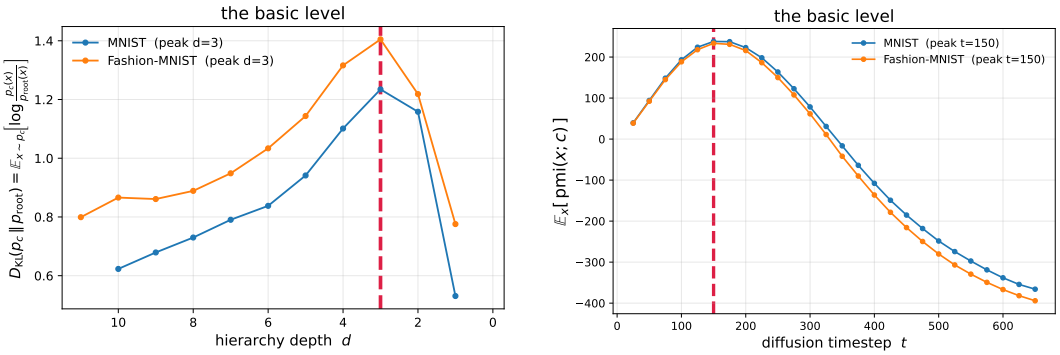


Figure 1. Concept hierarchies recovered by Cobweb (rows 1 and 3) and by score-based mode-finding in a diffusion model (rows 2 and 4), on MNIST and Fashion-MNIST. The Cobweb rows show its concept taxonomy, with the root at the top, prototypes at internal nodes, and exemplars at the leaves. The diffusion rows show the taxonomy recovered by the merge-path procedure of Section 4. Modes found at each noise level are grouped into prototypes that merge into coarser parents as the noise level rises. Panel heights are normalized for display, and lower levels are shown only in part, since the branching factor grows toward the leaves. The two models recover structurally similar trees.



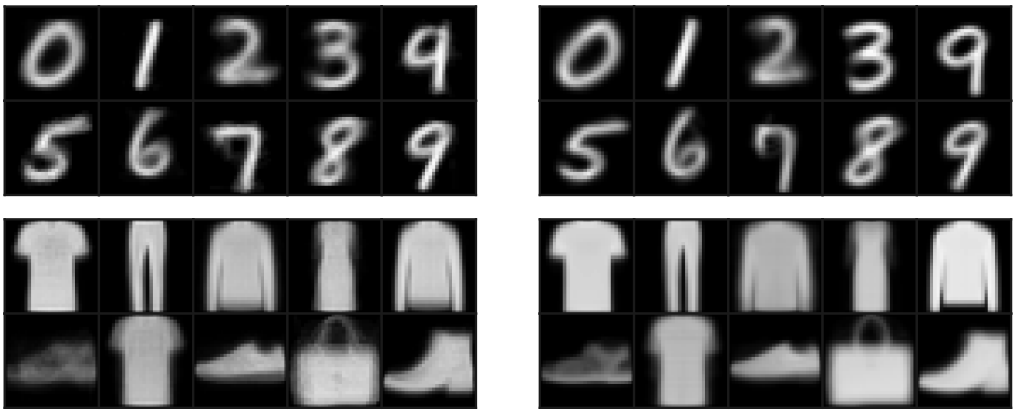
(a) Cobweb (b) Diffusion

Figure 2. Basic-level curves for the two models, each computed over both datasets. We plot the expected pointwise mutual information $\mathcal{D}$ (Eq. 11) against the level of abstraction, which is tree depth in (a) and noise level in (b). **(a)** In Cobweb, $\mathcal{D}$ against tree depth, peaking at depth three, with the root on the left. **(b)** In the diffusion model, $\mathcal{D}$ estimated along the mode paths against diffusion timestep, peaking near timestep 150. Each model shows a single interior maximum, the basic level (dashed line), rather than a maximum at the root or the leaves, on both MNIST and Fashion-MNIST.

and read the basic level off the level at which $\mathcal{D}$ peaks. In Cobweb the concepts are explicit, so we evaluate $\mathcal{D}$ at each node of the learned tree, approximating the data marginal by a mixture over the nodes. In the diffusion model the concepts are implicit, so we recover them by routing the held-out test images through the trained model along each datum's mode path (Eq. 14) and aggregate, giving $\mathcal{D}$ as a function of the noise level.

Figure 2 shows the resulting curves. Each has a single interior maximum rather than a peak at the root or the leaves. The Cobweb score peaks at depth three, and the diffusion score near timestep 150, for both MNIST and Fashion-MNIST. The diffusion peak falls at an intermediate noise level, consistent with the phase transition at which the reverse process commits to a sample's class (Sclocchi et al., 2025). Reading the concepts off these peaks makes the correspondence concrete. Figure 3 shows the basic-level prototypes of each model. These are the Cobweb nodes at depth three and the diffusion modes of the marginal at the peak timestep, recovered through Tweedie's formula. Both are object-level categories, coarser than individual exemplars yet finer than the dataset as a whole, and they align with the categories people treat as basic.

In addition to showcasing this qualitative similarity, we conduct a quantitative evaluation of the similarity of the basic level in both Cobweb and diffusion models, showing results in Figure 4. Utilizing a greedy matching scheme of basic-level concepts across the two models, we ask whether a Cobweb concept and its diffusion counterpart are the same concept. The diffusion frontier at $t^\star$ is the finer of the two (150 modes against 93 Cobweb concepts on MNIST, 108 against 64 on Fashion-MNIST), so we repeatedly pair the closest unmatched Cobweb concept and diffusion mode by prototype distance until every Cobweb concept has its best partner, then fold each remaining mode into the group of its nearest partner and merge the group into one concept. Each concept induces a distribution over ground-truth classes, and Figure 4 reports the KL divergence from each Cobweb



(a) Cobweb (b) Diffusion

Figure 3. The basic-level concepts at the peak levels of Figure 2. For each model we show ten prototypes from its basic level, one per class, with MNIST on the top rows and Fashion-MNIST on the bottom. **(a)** The Cobweb prototypes at depth three. **(b)** The diffusion prototypes recovered as the modes of the marginal at the peak timestep $t^\star \approx 150$ through Tweedie's formula. Each node is coarser than individual exemplars but finer than the whole dataset, and the matched pairs show the similarity of the basic level across the two models.

concept's distribution to its matched mode's vs. when the pairing is shuffled. Matched pairs diverge by 0.45 nats on MNIST and 0.42 on Fashion-MNIST, and by 0.36 and 0.28 when concepts are weighted by size, against 1.5 to 2.1 nats under shuffling.

Together, these results support our central claim that diffusion models and Cobweb belong to the same family of models, recovering the same concept hierarchy and the same basic level on MNIST and Fashion-MNIST.

5. Discussion and Conclusion

We have argued that diffusion models and Cobweb belong to the same family of models. Both are hierarchical density models, with tree depth in Cobweb playing the role that noise level plays in diffusion. Their concepts are Gaussian prototypes under a hierarchical Bayesian prior, stored at tree nodes in one case and read off the modes of the smoothed marginals in the other. Each categorizes by following a score toward the concept that most reduces uncertainty about the instance. And in each, a basic level emerges at which categorization takes the least cognitive effort, which we locate for a diffusion model at the intermediate noise level where the reverse process commits to a sample's class. Section 4 tests these claims. We recover the implicit hierarchy by mode-finding, then compare its shape and its basic level against Cobweb's, and the two align on MNIST and Fashion-MNIST.

The correspondence has two implications for the cognitive science community. First, diffusion models offer cognitive science a bridge to deep learning. Cognitive models of categorization have mostly been developed on small, hand-coded domains, while modern generative models train on natural images at scale but are treated as black boxes. The concept formation reading connects the

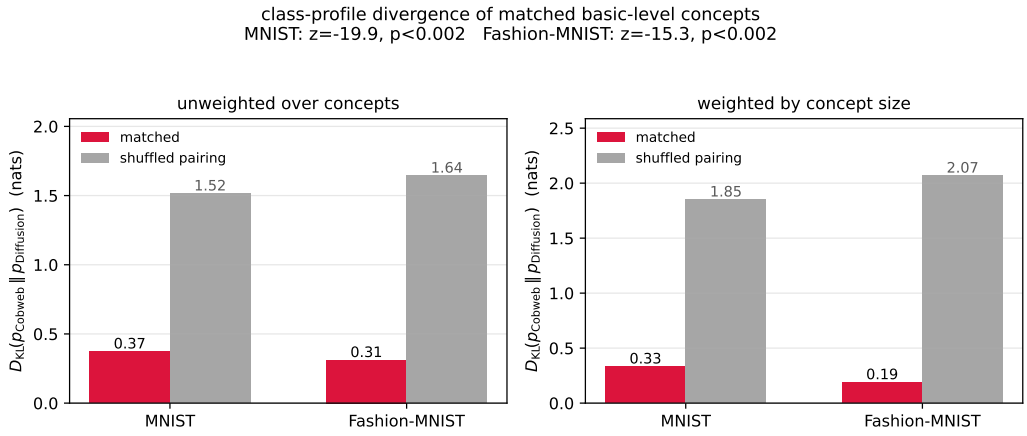


Figure 4. Class-profile divergence of matched basic-level concepts. Kullback–Leibler divergence from each Cobweb basic concept's distribution over ground-truth classes to that of its greedily matched diffusion mode, **(a)** averaged over concepts and **(b)** weighted by concept size, against the same statistic when the pairing is shuffled. Matched concepts diverge three to seven times less than chance on both MNIST and Fashion-MNIST.

two. Prototypes, graded typicality, and the basic level all acquire precise computational analogs inside a pretrained diffusion model, namely the modes and the noise level of maximal distinctiveness. Hypotheses about human categorization can therefore be posed and tested on naturalistic stimuli without building a bespoke cognitive architecture, and the results return to the machine learning side as cognitive interpretations of what these models have learned.

Second, this study is a proof of concept for studying symbol acquisition in neural models. With diffusion models, mode-finding extracts discrete, stable prototypes from a continuous density, and these proto-symbols compose, in that combining their densities yields instances that satisfy several concepts at once (Wang et al., 2026; Du et al., 2020; Liu et al., 2022). Text-conditioned diffusion models already pair such densities with words (Ho & Salimans, 2022). A natural next study is to ask how a word comes to pick out a mode of the density, at which level of the hierarchy words attach most readily, and whether that level is the basic level identified here. Broadly, symbolic and neural systems can be combined in many ways (Kautz, 2022), and the correspondence we draw shows one concrete route, pairing a symbolic taxonomy with a neural density model.

The limitation of our approach lies in how the hierarchy is indexed. We inherit the noise schedule of diffusion practice, typically 1000 uniformly spaced timesteps whose spacing was tuned for sample quality rather than for cognitive plausibility (Ho et al., 2020; Karras et al., 2022). Equal steps in timestep are not equal steps in abstraction. The mapping from timestep to signal-to-noise ratio is nonlinear (Kingma et al., 2021), so the recovered hierarchy resolves some ranges of abstraction finely and others coarsely, and the basic level is reported in units that carry no cognitive meaning of their own. Future work should explore cognitively inspired schedules that place noise levels at human gradations of abstraction, for example calibrating successive levels to subordinate, basic, and

superordinate structure (Rosch et al., 1976), or spacing them by equal increments of category utility rather than of time. Indexing the hierarchy by level of abstraction rather than by timestep would make the correspondence with human concept hierarchies exact in form as well as in structure.

Acknowledgements

We would like to thank Pat Langley for discussions.

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